AtlasSAT-16U Souss Water Observation CubeSat

Independent Research Project | Document 2A | Orbital Analysis | Rev D3.4
wassim.aitoubahou@gmail.com

16U CubeSat · 550 km near-SSO · GMAT R2026a
VNIR pushbroom · 30 km nominal swath
Chtouka-Massa agriculture, south of Agadir

ENGINEERING OBJECTIVE

Test whether one AtlasSAT-16U can offer agricultural targets south of Agadir a **daylight opportunity every** ≤ 72 **h**. Preserved GMAT dynamics, provisional GIS targets and checked optical-access calculations form the evidence.

PRESERVED CONCEPT

Model	Adopted input
Orbit	Near-SSO, 550 km concept
Semimajor axis	6928.1363 km
Ecc. / incl.	e=0.001; i=97.5°
RAAN	6.463°
AOP / TA	0° / 0° (canonical)
Epoch (UTC)	20 Oct 2026, 12:00:00.000
Initial axes	EarthMeanOfDate (Earth; MODEq)
LTDN / span	10:30 mean solar; 30 days
Payload	Five-band VNIR pushbroom
Swath / roll	30 km; nadir, ideal $\pm 10°$ and $\pm 20°$
Daylight	Target Sun elevation $\geq 0°$

WHAT WAS VERIFIED

Archived orbit benchmark: **36 numerical checks**. Coverage geometry: **570 opportunities at 31 targets**; maximum boundary difference **1.197** μ**s**. Numerical consistency is not physical accuracy.

FOUR EVIDENCE LEVELS

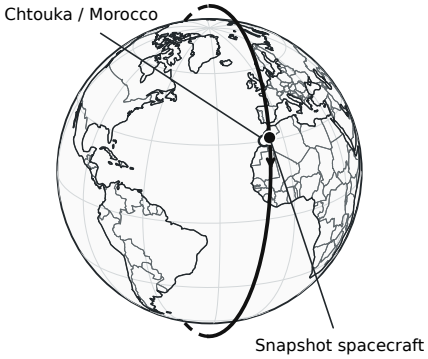
Passage: a track crosses the region.

Reachability: a target lies inside an ideal sensor envelope.

Daylight opportunity: reachability plus illumination.

Physical validation: an achievable imaging service remains unproven.

PRESERVED NEAR-SSO GEOMETRY



1 Nov 2026, 11:23:49 UTC snapshot. True orbit-to-Earth scale; solid near-side / dashed far-side arc. Predicted orientation. Natural Earth context.

CHTOUKA–MASSA AGRICULTURAL TARGET



Seven-commune study boundary
Primary agricultural mask (provisional)
Major roads (OSM)
Greenhouse-source southern edge

*Morocco locator: geographic context only.
WGS84 / UTM 29N; grid-north arrow.

396.47 km^2 primary mask in a **1,173.92 km**^{2} seven-commune envelope. **Provisional research ROI**, not an official agricultural perimeter.

KEY RESULTS | DAYLIGHT | 30 km

Metric	Nadir	$\pm 10°$	$\pm 20°$
Mean 72 h area	11.48%	22.76%	33.44%
Max complete gap	360.01 h	360.01 h	**312.22 h**
Largest span*	720 h	360.01 h	312.22 h

*Includes endpoint censoring. Hourly window starts; 250 m area quadrature. Opportunities, not acquired images.

ENGINEERING SIGNIFICANCE

Slow ground-track drift clusters opportunities rather than providing daily service. A favourable mean cannot replace a maximum-gap test; a night pass cannot meet daylight needs. The analytical note checks this interpretation; future design trades remain unperformed.

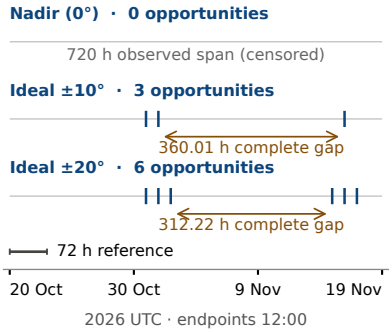
DAYLIGHT REVISIT VERDICT

NOT MET

312.22 h vs **72 h**

Longest complete gap with ideal $\pm 20°$ roll. All 14,494 samples fail the 72 h test.

30-DAY DAYLIGHT ACCESS TIMELINE



30 km; Sun $\geq 0°$. Ticks: point opportunities, not acquired images or time-scaled bars. E16.

TRACEABLE WORST CASE

Sidi Bibi target `sidi-bibi_-10750_19500`; 2–15 November. REF lies outside the ROI and is **excluded**. Full coordinates and UTC endpoints: 2A-1 / E16.

BOUNDARIES OF THE EVIDENCE

- **Provisional** agriculture; local classification accuracy unestablished.
- Greenhouse-source gap: **22.04%** of envelope / **6.861%** of mask. Missing acreage unknown.
- **Predicted** Earth orientation; preliminary forces.
- Ideal roll is not qualified slew, pointing or image quality; physical/operational acquisition unvalidated.
- TA90 changes initial TA at fixed osculating inputs, not phase alone at fixed mean orbit.

NEXT STEPS

Next: phased 2–5 satellite constellation trades, followed by orbit-design, drag, solar-elevation and observation-scheduling studies.

DETAILED ENGINEERING RECORD

2A-1, Rev D3.4: methods and gap provenance. E15: native GUI-first script (bundled). Codex-assisted implementation; detailed contribution note supplied.

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